

Unveiling the Media Landscape: A Data Analysis of 2024 U.S. Election Coverage

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Abstract

In the wake of growing political polarization in the United States and the rapid evolution of technology, particularly in the realms of social media and artificial intelligence, the question of media bias in news coverage of political parties is more pertinent than ever. This report seeks to examine whether American news outlets exhibit bias in their coverage of the 2024 presidential election across political lines. In particular, this report aims to highlight trends concerning the two primary candidates, Donald Trump and Joe Biden.

Initial exploratory data analysis (EDA) was conducted on a collection of recent news articles, categorized into four groups: Trump, Biden, Election, and Policy, using keywords related to the 2024 U.S. Election. We gathered our article data using an API called NewsAPI and used Mongo as our NoSQL database of choice. The subsequent analysis involved creating visualizations such as pie charts, word clouds, and histograms to depict the distribution of article categories, common vocabulary, source distribution, character counts, and publication dates.

Most news articles were not extremely scored and had a borderline neutral sentiment score. The core of the analysis focused on the sentiment scores of articles differentiated by the political viewpoint: left-leaning, right-leaning, and centrist news sources. Centrist news sources act as a baseline of how less politically affiliated news sources should convey their findings. Graphs revealed a very similar negative average sentiment score across both left-leaning sources and right-leaning counterparts. Centrist scores were scored around the same magnitude but in a positive direction. Often these averages were skewed by more extremely scored sources within these classifications. Detailed sentiment comparisons between specific news outlets on specific categories failed to depict strong trends from expected differing viewpoints on categories like President Biden or the election in general. The specific findings between sources indicate nuanced differences in how news outlets cover political figures and election-related topics. ANOVA analysis depicted that there were no significant differences among the average scores of sources within each political viewpoint.

We concluded that these sentiment score differences amongst politically-affiliated sources were not significant, and therefore, media bias on election was not detected on partisan lines. Some limitations of our analysis include limited content to score these articles and sentiment score's inability to capture subjective biases.

Introduction

As data science students at a top American research university, we are acutely aware of the rapid speed of technological innovation. These advancements have undoubtedly improved modern society in many ways, one of which is facilitating a faster and more accessible flow of

information. With the upcoming American presidential election approaching, our group was particularly interested in understanding how news coverage from mainstream outlets differs between the Democratic and Republican party candidates (namely Donald Trump and Joe Biden). This issue was of special importance to our group. We have grown up in an especially polarizing time in American politics; the previous 2 elections (2016 & 2020) occurred during our late middle school/ high school years. With added factors like misinformation on apps like Facebook and TikTok and ever-growing applications of Artificial Intelligence, it is now more important than ever to get news from a trusted source. Now, as collegiate data scientists with skills that enable us to probe deeper into this issue, our *driving question* for our project was, “Are American news outlets using titles and content that propagate polarization amongst the two parties for the 2024 presidential election?” Additionally, we wanted to see if our news headlines data showed any *specific patterns* amongst news outlets (or coverage on the two parties), particularly relating to bias.

Methods

For our project we chose the Application route with Mongo as our NoSQL database of choice. The first step was to gather JSON data with 2024 election news headlines. For this, we found the [News API](#) that scrapes news article information from the web. Using a free trial plan, we obtained an API key and got to work. An important note is that we could only obtain data *from the past month* with the free plan; we found this acceptable, as there are a plethora of recent news articles discussing the upcoming election. We utilized the “get_everything” endpoint

To obtain the specific data we wanted, we created a Jupyter Notebook called ‘HW 6 Data Cleaning File’ with 2 functionalities: 1. Gather news article data via the News API and 2. Data cleaning. We filtered the articles by determining if 3 attributes of a given article had our predetermined keywords: 1. The article headline (title), 2. The description of the article 3. The content of the article. We used these keywords to search using the API endpoints: '2024 U.S. Election', 'Primary', 'President', 'Biden', 'Trump', "Republican", "Democrat", 'GOP', "Vote", and "Ballot". The data was gathered based on a downloaded python module that the API had created to return results. Based on this data, long strings were split, character counts of the total articles were calculated and sentiment scores were calculated using the compound score from the nltk sentiment vader module. These sentiment scores range from -1 to 1 and represent if the overall description was overwhelmingly negative (closer to -1), neutral (close to 0), or overwhelmingly positive (closer to +1). Additionally, using nltk stemmer module, an article is then given one of four categories: Trump, Biden, Election, and Policy (or “Other U.S. News” if it does not fit any of the 4 categories) based on the article’s description keywords. Each category contains a list of associated keywords, i.e. for the ‘Election’ keyword, an example of keywords are: "U.S. 2024 election", "vote", '2024', 'primaries', 'u.s.', 'ballot', and 'campaign'. The ultimate result of ‘HW 6 Data Cleaning File’ gives us a JSON file called ‘final_data.json’, which contains the title, news

source, description, content, category, sentiment, date, and link of the respective articles. There were a total of 6018 articles in our data from 512 different sources.

Next, the file ‘news_api.py’ is where we implemented a NewsAPI class that is used to query data in Mongo. It contains several functions that allows the user to gather and aggregate sentiment scores utilizing the power of Mongo queries in various ways and effectively visualize the findings. The three stages of our analysis consisted of: Exploratory Data Analysis, Understanding the Distribution of Sentiment Scores, and ANOVA statistical testing. Exploratory Data Analysis consists of visualization functions to better understand different attributes of the data. One example is the function ‘get_top_words’, which finds the highest counts of words among all of the news data and visualizes the findings in a word cloud. The next stage consists of visualizing sentiment scores for all sources, and then differentiated on political lines. We classified well-known sources from the data based on their known political bias or affiliation using online resources (i.e. <https://www.allsides.com/media-bias/media-bias-chart>). We were unable to classify all of the extensive number of sources for their political affiliation so we narrowed our scope. The “right-wing” sources include "FOX News", "The Daily Caller", "Breitbart News", "The Blaze", "Washington Free Beacon", "The American Conservative", "The Daily Caller", "The Federalist", "Americanthinker.com". The “left-wing” sources include "CNN", "HuffPost", "MSNBC", "The Washington Post", "Democracy Now!", "The Atlantic". The “centrist” sources include "BBC News", "PBS", "Forbes", "Associated Press", "Pew Research Center." Functions in this analysis plotted histograms and bar plots of average to get a better sense of trends, along with pairwise comparison to see source views on specific categories. Another example of a specific is the function ‘sentiment’, which returns the sentiment score aggregated for a specific list of sources and the possibility for averaging the values using Mongo. The last stage of our analysis was to determine statistically whether there was a difference in average sentiment scores among sources in the three political affiliations: right-wing, left-wing, and centrists using ANOVA. This was to enable whether there is an actual difference in attitudes of these sources based on media bias.

Finally, we have the file ‘HW 6 Analysis’ which calls the functions from ‘news_api.py’ and displays our visualizations and shows final results. We will discuss our findings in the Analysis section, the following part of our paper.

Analysis

Preliminary EDA

The first part of our analysis will consist of drawing insights from the preliminary EDA we performed on our JSON news article data. We created 5 EDA visualizations, with the goal in mind of understanding the *distribution* of our data better.

Figure A is a pie chart that displays the proportion of articles that belong to each of the 5 broad categories that we categorized the news articles into. It was created from our API function ‘topic_analyzer’. The pie chart shows us that the category with the highest number of articles is “Other U.S. news” (40%), followed by ‘Trump (24%), and ‘Election (~17%). We found it quite interesting that there are 14% more articles in the ‘Trump’ category than the ‘Biden’ category. It is suspected that this is due to Trump’s impending criminal trial, which has consistently made headlines for the past month. However, knowing the overlap between these categories, categorizing is a difficult task to achieve correctly.

Figure B is a word cloud that displays the most common words found amongst all of the articles, created with ‘get_top_words’. Although this visualization is quite simple, we felt that it was important to understand the most used words amongst the articles. ‘President’, ‘former’, ‘Donald’, and ‘Trump’ were amongst the most popular, which suggests many of the articles have to do with Donald Trump’s criminal trial.

Figure C displays the distribution of the news outlet sources that the articles were published by. We found this visualization to be particularly insightful, as different outlets have variations in things like political affiliation, so it was important for us to understand which sources were being represented the most and least. As can be seen from the visualization, the top 4 sources in order are: Business Insider, ABC News, Forbes, and Biztoc.com. 3 out of 4 of these are business sources meaning many of the articles come from business perspectives. The bottom 6 sources are as follows: Marketscreener.com, BBC News, NPR, The News Republic, Daily Beast, and Time. Notably, Daily Beast is characterized as a “high-end tabloid”, which suggests its headlines and articles could be more “dramatic” than traditional news sources, which is important when analyzing articles from this source. Overall, a majority of the news sources in our data come from business news sites, with the most of the remainder being more “traditional” news reporting outlets (i.e. ABC News and Time). This means that much of our data analysis will come from a “business” perspective.

Figure D displays the distribution of the number of characters in each article, using ‘character_count’. The results here showed the most common character count of the articles we gathered with our API consisted of 0-2,500 characters (1,600 articles). Most articles have a character count between 0 - 6,250 with many on the shorter side of this range seen by the right-skew. So, we were able to conclude that from the election/political news articles we collected, many have a relatively short character count. This aligned with our intuition that political informational news articles (like we are analyzing) may be less wordy compared to other types of online media. Around 77 articles were not included in the visualization as they significantly intensified right-skew as there were articles outside 40000 characters. This indicates that despite the majority of the articles, few were extreme outliers.

Figure E represents the distribution of dates the respective articles were published. The aim of this visualization was to try to get a sense of temporal trends of specific election topics. Perhaps spikes from certain breaking news would be represented from the number of articles that day. However, the graph is characterized by an outlier data point on the upper right corner of the graph, which indicates the most common publish date is '2024-04-01', with 1,750 articles. Many of the articles we will be analyzing will be quite recent. This reflects correctly on the orientation of the API which updates continually and provides the most recent news articles as our data was sourced around that same date. Therefore, there are no strong findings in regards to temporal trends.

All News Sources

The second overarching component of our analysis was the 'All News Sources' section, where we analyze the sentiment score of articles amongst all of the respective sources (both left-leaning and right-leaning sources). Figure F displays a histogram that shows the distribution of the articles' sentiment scores. Much to our surprise, we noticed that the distribution had a higher number of articles with a positive sentiment score than a negative one. The sentiment score bin with the highest number of articles was 0.0-0.25 (1,600 articles). While this bin contains many articles with a "neutral" sentiment score (around 0), it can be seen that ~600 articles have a score between 0.25-0.5 and ~700 with a score between 0.5-0.75. However, only ~300 articles had a score between 0.0 and -0.25. The histogram was useful in visually depicting that slightly more articles have a positive sentiment, as seen by the higher bin article counts on the right side of the x-axis (positive values). This went against our previous assumption that a majority of the articles would have a negative sentiment score.

Media Bias (left vs. right)

The third portion of our analysis (and the most critical) examined the articles split into left-leaning sources vs. right leaning sources. Before we began analyzing specific articles affiliated with either political side, we chose to take a broader look at the average sentiment scores across the political divide. Figure I represents the average sentiment across the articles based on political bias. The scores in this figure show that while both left-wing and right-wing media have an average negative score around ~ -0.27 . This already depicts that there is not a significant difference between the two opposite political views based on sentiment scores. However, knowing that averages are sometimes skewed based on the data distribution, we plan to investigate distributions later. Centrist sources serve as a means of comparison and the average is around the same sentiment score in the complete other direction, positive. This might indicate that these articles from centrist sources tend to use more positive language to convey their message or are trying to avoid critical statements on the current election cycle to maintain more neutrality.

Looking at distribution of sentiment scores in Figure G based on left-wing and right-wing sources, the ranges are very similar. It is clear that both political sources are concentrated around more neutral sentiments with the left-wing media highest bin at 0-0.25, and the right-wing media at -.25-0.

In Figure H, we solely looked at the sentiment scores of centrist-leaning sources. There is a clear spike around 0, suggesting an overall net-neutral sentiment. Given what we know about centrist affiliation and their lack of predominant bias to either primary candidate, the spike at 0 is not a surprise, nor does it deter from our hypothesis.

Figure N serves to compare average sentiment scores across sources known for bias. Heavy variance is present on both the left and right-leaning sources. This holds with the prior takeaways that there is not a significant difference between the opposite political views in terms of sentiment scores. Additionally, the skewing of the scores is most likely due to a heavily weighted sources.

Figure O depicts the average sentiment scores across centrist sources. In comparison to the overall average of centrist sources that was relatively positive, this depicts a better understanding why. Associated Press seems to be skewing the average to be artificially high with most likely a larger number of positively scored articles. The other sources indicate a more neutral, or not as significantly, positively scored averages. A more neutral stance would be expected from centrist sources as they try to avoid politically extreme perspectives.

With this baseline established, we began to look at the contrast of specific sources. Figure J shows the distribution of the negative and positive sentiments of individual articles by Breitbart News (right-biased) and ABC News (left-biased) in the election category. Overall, while both sources seem to have a large cluster around the 0 (neutral) mark, ABC News tends to have a higher percentage of total articles with a more positive sentiment in comparison to Breitbart News, which seems to hold with the analysis we completed with the larger sentiment analysis. It is worth keeping in mind for this analysis, that there are significantly more data points for ABC News in comparison to Breitbart News. Based on conveying information or updates on the election, it makes sense that articles would have a more neutral tone. Figure K highlights the differences of sentiments between Breitbart News and BBC News in the Biden Category. While the overarching analysis was done for left-biased and right-biased sources, we felt it was important to include the difference in sentiments from center-biased sources as well, which in this case is BBC News. Looking at the figure, BBC tends to have articles that are concentrated pretty heavily around the 0 or neutral mark, which holds with what we know about centrally-biased sources. Especially, as a news source detached from American politics based in the U.K., their neutrality on Biden is in line with our expectation. On the other hand, Breitbart News has a larger range of scores, with a more substantial amount dipping into the negatives. As

a right-wing source, expressing a more negative take on Biden is consistent with American political polarization. Figure L analyzes the difference in sentiments between two centrist sources, with NPR and BBC News, based on the election category. As shown, they both tend to have a wider range of sentiments in comparison to BBC News. This mirrors what we've seen in previous analysis and holds with what we know about center-biased sources — they tend to be more neutral than those that are politically charged either left or right. Overall, the BBC articles tend to be less skewed to the negative sentiment end in comparison to NPR, which does have quite a few negative scores. NPR has been sometimes considered politically biased to the left. Their negative scores from articles on the election could be from this potential bias. In Figure M, we analyzed yet another comparison between right and center-biased sources (The Daily Caller and BBC News) based on the Biden category. The Daily Caller has a tendency to have scores below the 0 mark, indicating a negative overall sentiment, amplifying the trend we've seen with right-biased articles, especially on a left-wing President. On the flip side, BBC News is predominantly clustered around the 0 mark, yet again showing the overall neutral scoring of center-biased articles.

ANOVA

The null hypothesis of our ANOVA testing was that there was no significant difference in the average sentiment scores of news sources between the three main political affiliations. The alternative hypothesis indicates the opposite, that there is a difference, which is what we had expected. Before conducting ANOVA, the assumptions of normality and homogeneity were assessed and passed. The F-statistic, the test statistic for this analysis, was reported to be ~ 0.188 . The p-value was reported to be $\sim .83$. Since the p-value is significantly higher than 0.05, we fail to reject the null hypothesis. This indicates that there are significant average sentiment scores between the news sources across political affiliations. This is substantiated by the boxplot that visualizes these results (*Figure P*). While the ranges significantly differ by each affiliation, most of the data for each affiliation clusters around 0, or neutral sentiment. The right-wing sources have the most range in average scores from relatively very negative to very positive, which was unexpected as we expected right-wing media to be more skewed towards negative scores. Left-wing media is the most neutral amongst its sources which was also surprising as left-media is often accused for its bias towards the current democratic administration. Centrist media ranged from neutral to positive sentiment range which is in line with expectations as Biden is considered a centrist president.

Conclusion

Despite our initial hypothesis that news coverage might significantly differ in sentiment based on political affiliation, thereby contributing to the polarization between Democratic and Republican party supporters, our analysis revealed no substantial disparities in the sentiment scores across

the political divide. While a minority of highly skewed sources did elevate the average sentiment scores, the majority of articles maintained a neutral stance, indicating a typically balanced representation. This was substantiated by multiple graphs and our ANOVA testing.

When conducting pairwise source comparisons, the results revealed a narrower range of sentiment scores among centrist sources compared to those we identified as right or left-wing. This suggests a more moderate tone in the coverage from centrist outlets. Additionally, one-to-one comparisons between right and left-biased sources indicated a tendency for right-wing outlets to publish more negatively scored articles. This could stem from the topics selected for analysis, and deeper analysis on these comparisons would be interesting to accomplish in our next steps. Overall, these observations offer more insight into the landscape of political news coverage and highlight the nuances in reporting that we wouldn't normally expect from such a starkly divided political spectrum. Additionally, these comparisons focused on specific categories. We would typically expect stark differences in reports of Biden from left vs right-wing sources, but no trends were depicted.

Despite our findings, our analysis did have limitations. As mentioned previously, our data's scope was set to the past month, which may not have allowed us to fully capture broader trends. Additionally, the API did not include the full content of the articles, which may have limited the extent of our sentiment scoring. Also, sentiment scoring may not be the most effective manner to conduct media bias detection.

Understanding media bias, specifically in terms of the highly polarized political landscape, has continued to be a difficult phenomenon to clearly distinguish. Even categorizing sources as left-biased or right-biased is a subjective matter which is a type of classification data analysis struggles with. For further analysis, we aim to scrape full article texts from news sites over a longer period of time, allowing for a more in-depth analysis of sentiments. Additionally, we see other valuable opportunities to expand our analysis in the future. Our current report predominantly focused on business-related articles, so it would be interesting to see how coverage and sentiment scores differ across different journalistic perspectives. Furthermore, splitting our labeled dataset of the three identified parties and applying text classification techniques could contribute further insights and uncover hidden biases in the verbiage of news coverage.

Appendix

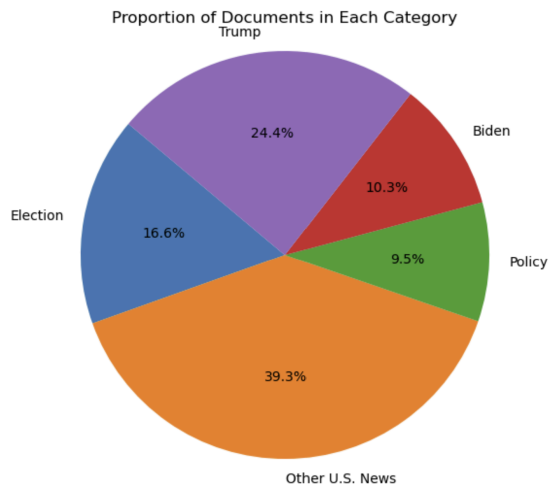


Figure A - 'Category Pie Chart'

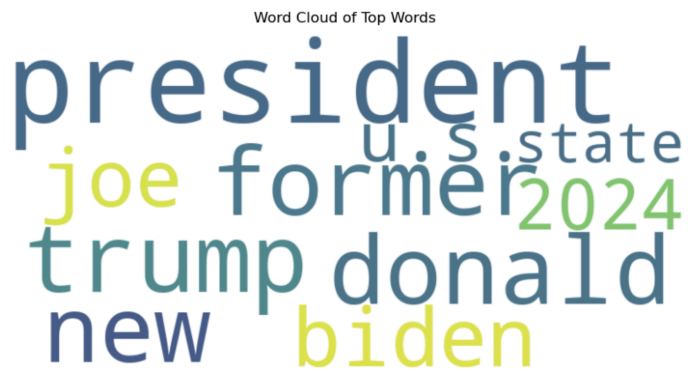


Figure B - 'Top Words Word Cloud'

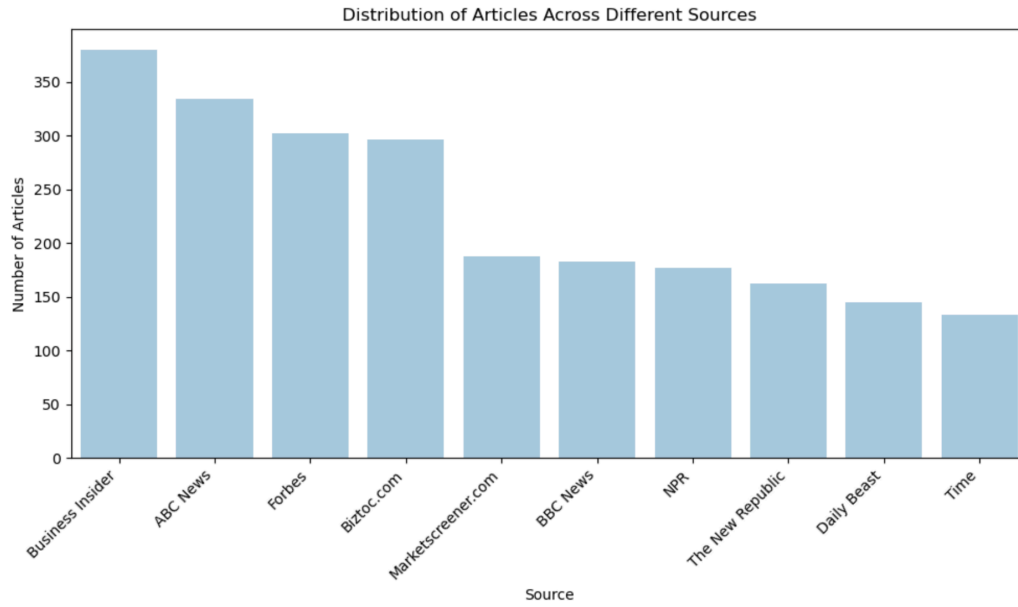


Figure C - 'Distribution of Articles Across Sources'

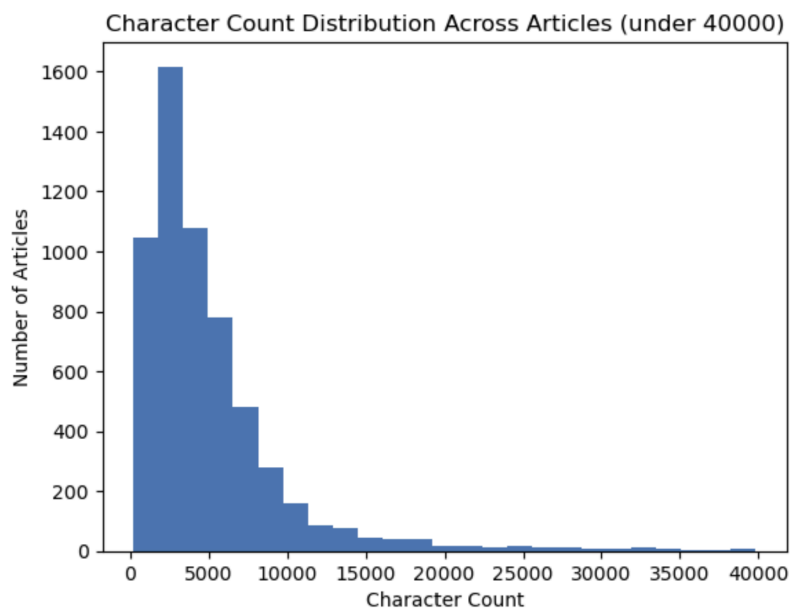


Figure D - 'Number of Characters in Articles'

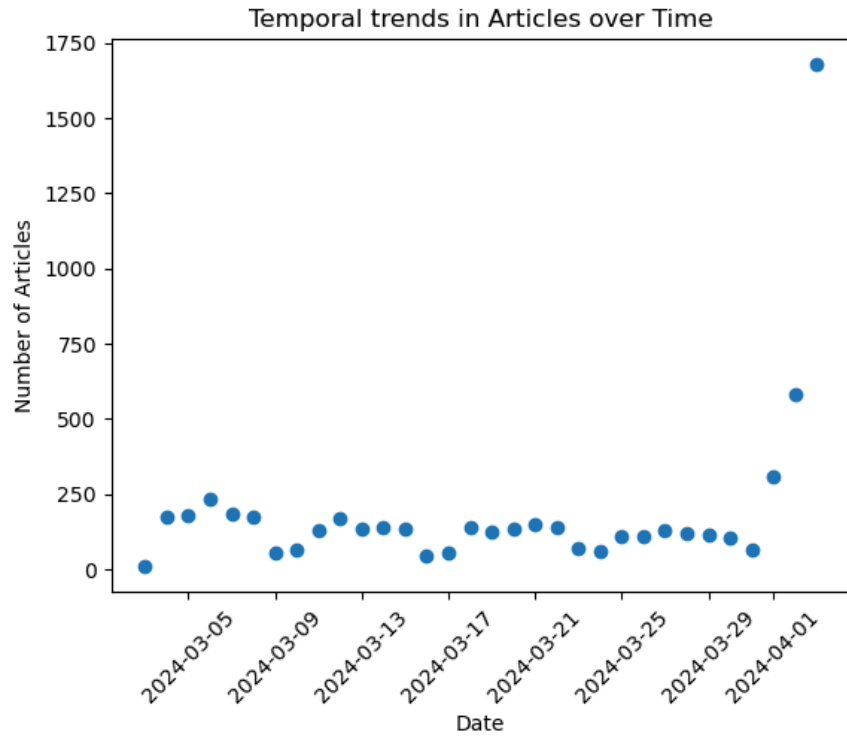


Figure E - 'Distribution of Articles by Publishing Date'

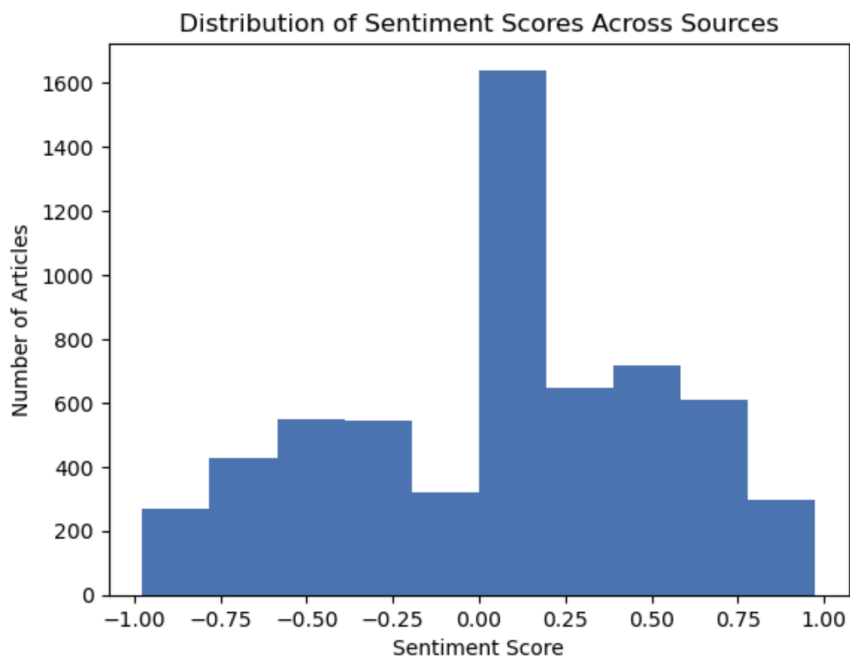


Figure F - 'Distribution of Sentiment Scores Across Sources'

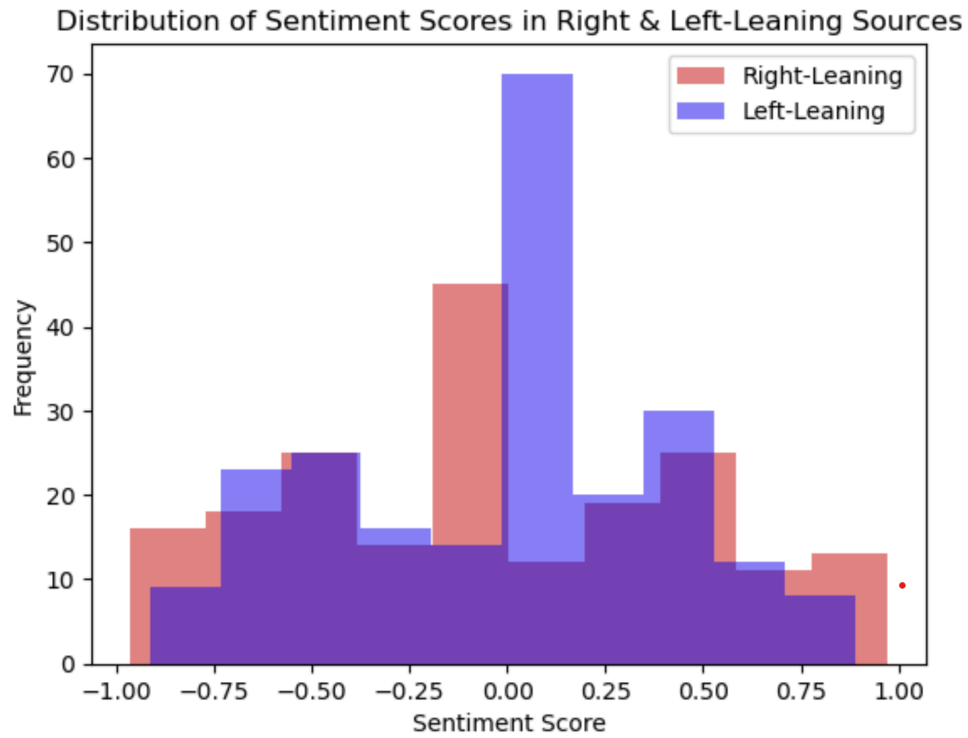


Figure G - 'Distribution of Sentiment Scores Across Right & Left Leaning Sources'

Figure H - 'Distribution of Sentiment Scores Across Centrist Sources'

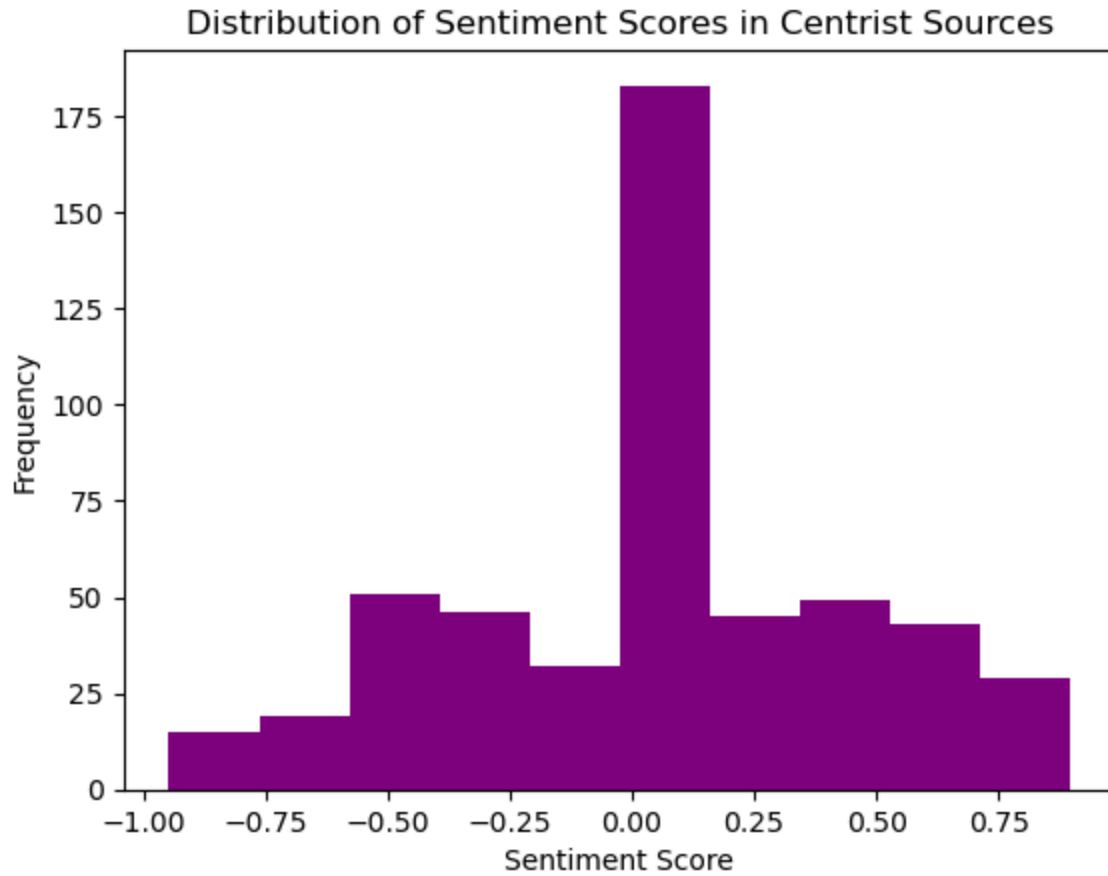
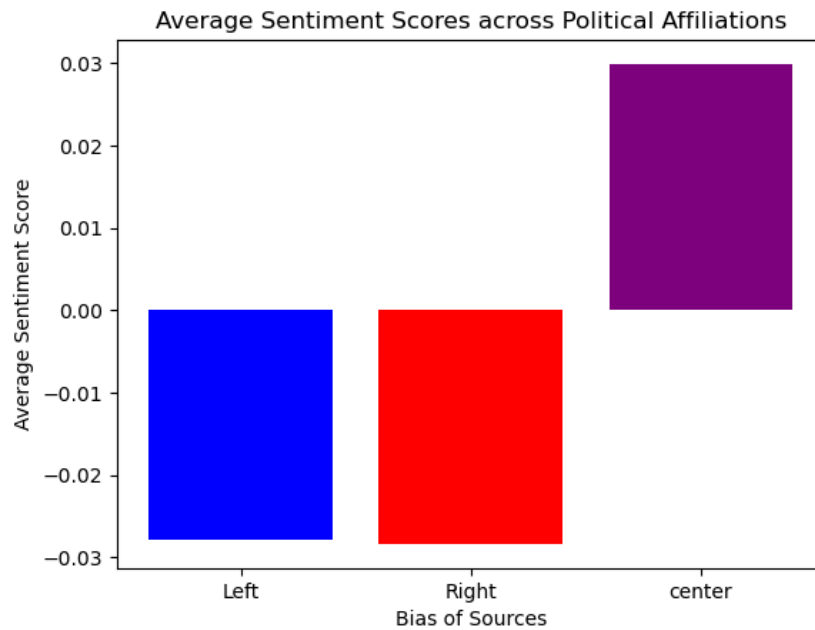


Figure I - 'Distribution of Average Sentiment Scores Across Political Affiliations'



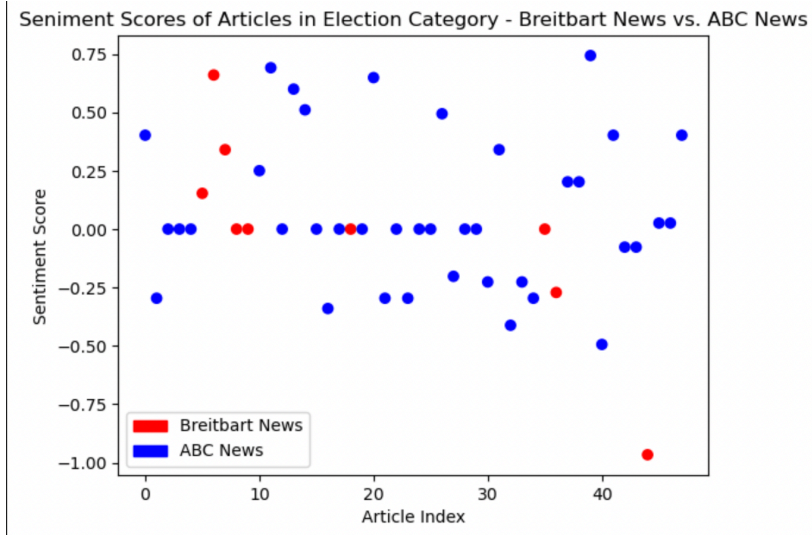


Figure J - 'Sentiment Scores of Breitbart News vs ABC News'

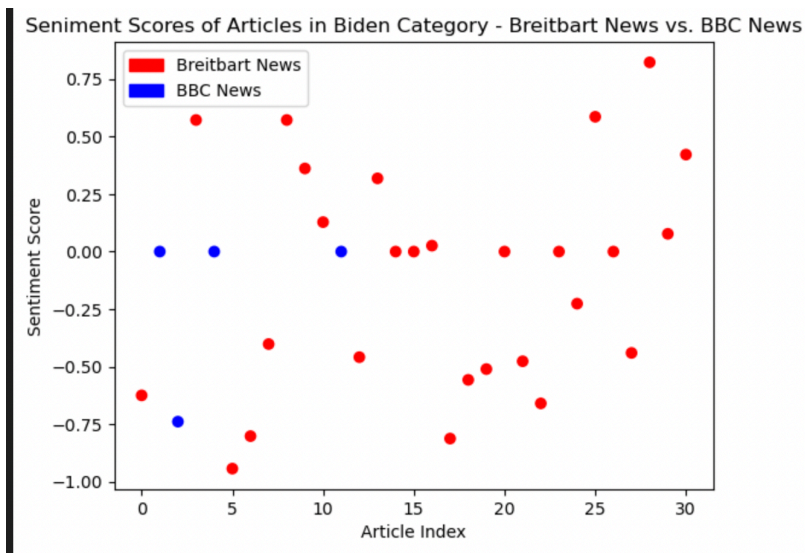


Figure K - 'Sentiment Scores of Breitbart News vs BBC News'

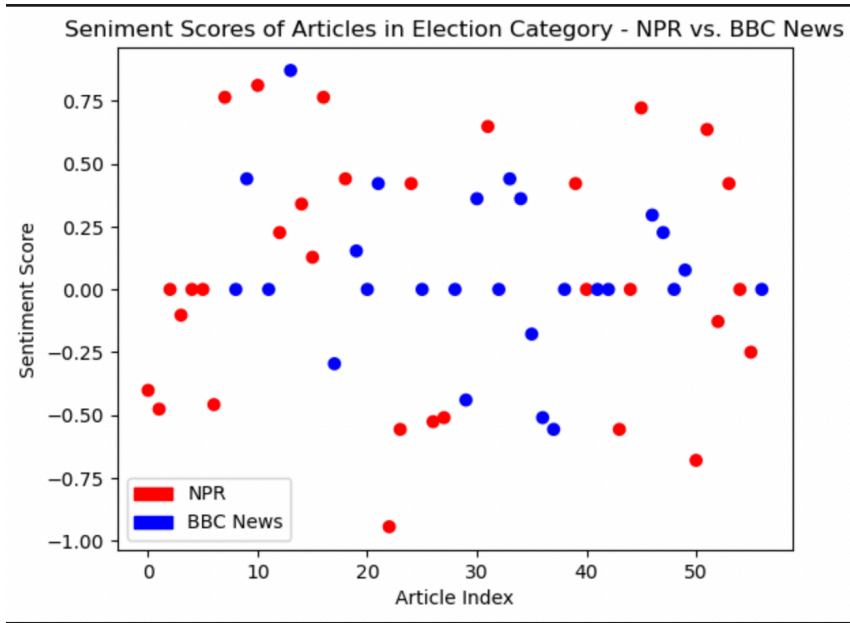


Figure L - 'Sentiment Scores of NPR vs BBC News'

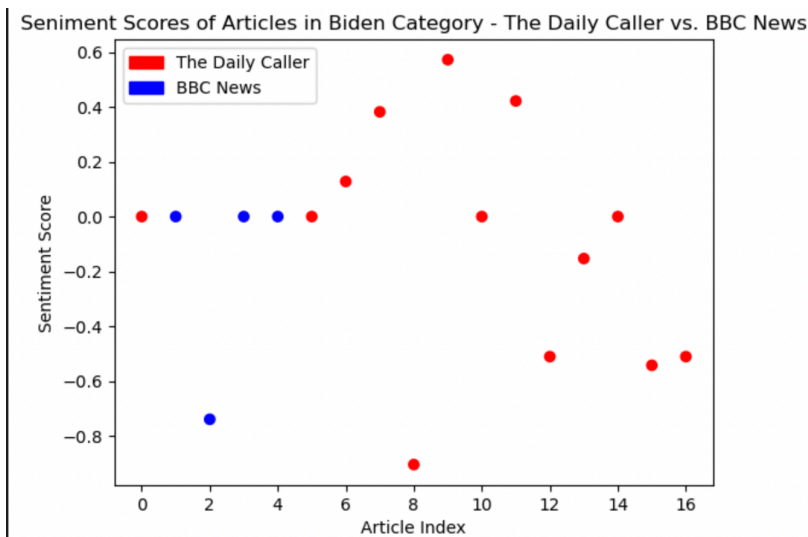


Figure M- 'Sentiment Scores of The Daily Caller vs BBC News'

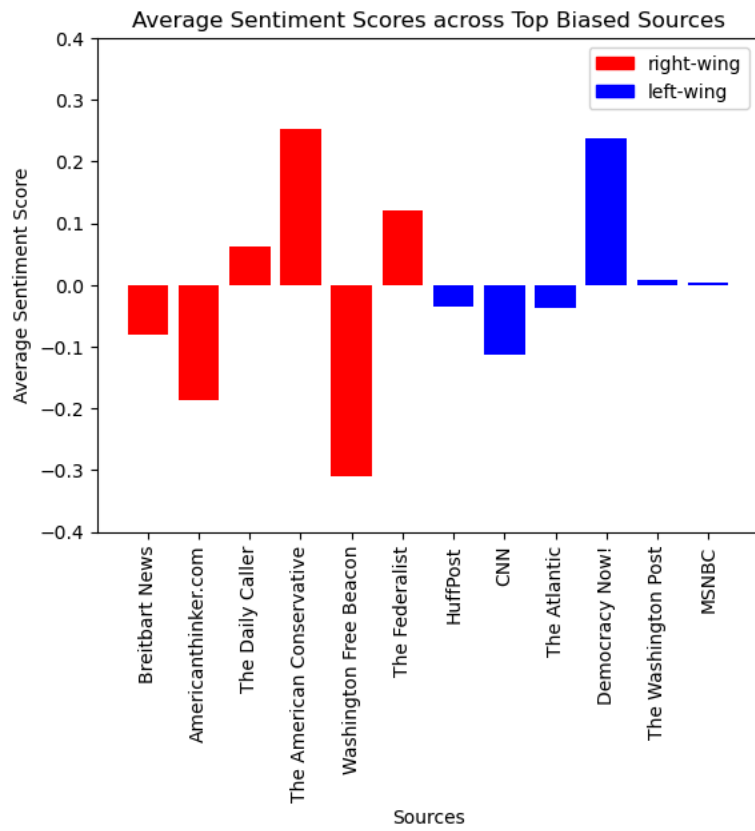


Figure N - 'Average Sentiment Scores Across Top Biased Sources'

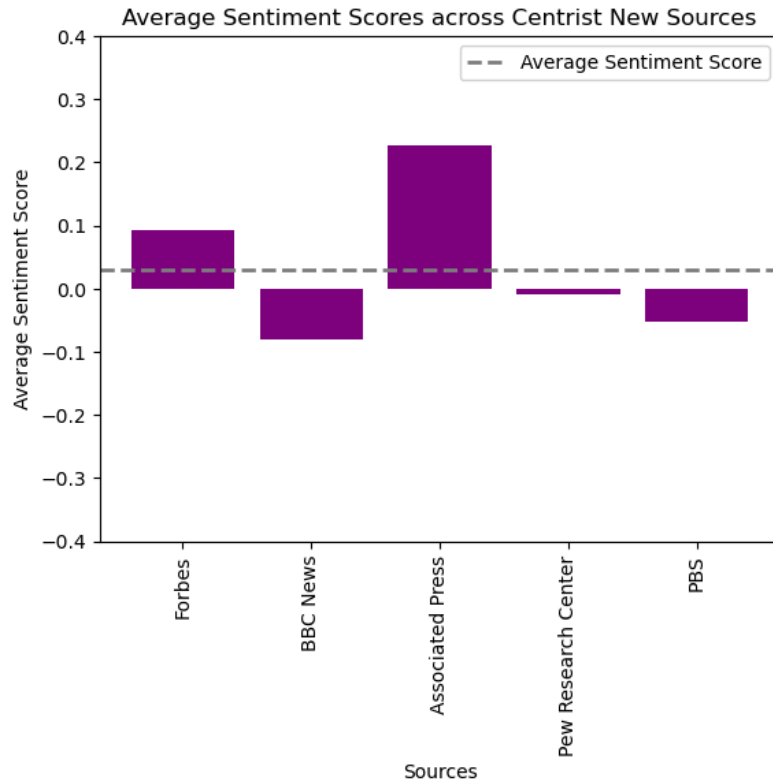


Figure O - 'Average Sentiment Scores Across Centrist New Sources'

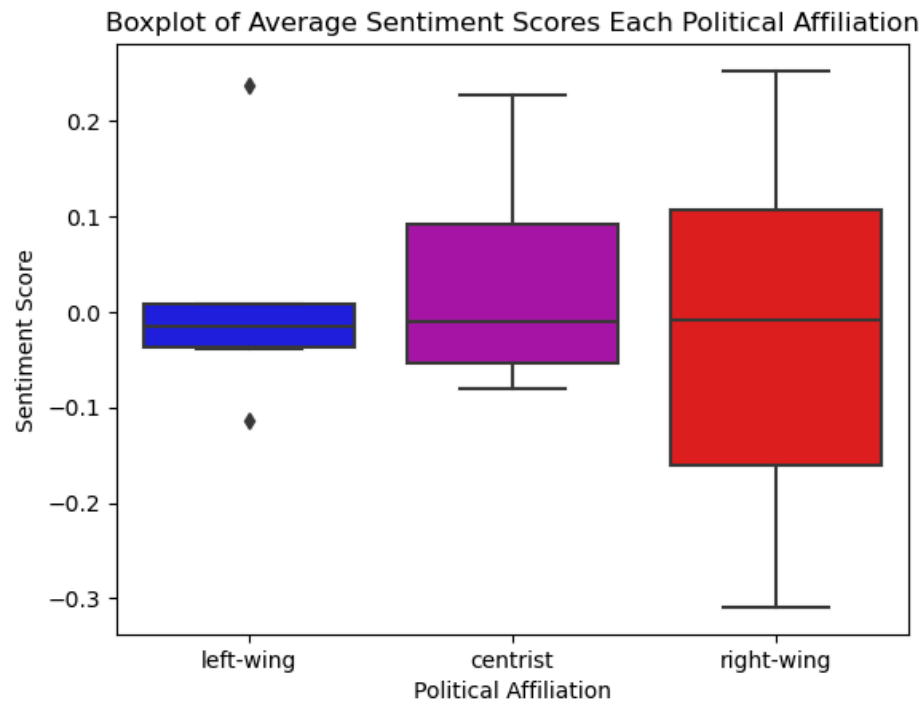


Figure P - 'Average Sentiment Scores Across Each Political Affiliation'